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Expected game value (xGV): a novel approach for strategic decision-making in rugby union

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ABSTRACT

This study introduces expected Game Value (xGV), a new analytical model for rugby union that quantifies the value of each game state. Unlike traditional methods relying on overall match data, xGV provides an intra-game perspective by weighting the impact of each action based upon its context. While comparable measures used in other sports often rely on machine learning methods, xGV employs generalised additive mixed models (GAMM), which account for the impact of teams as random effects while maintaining parameter flexibility through the use of penalised splines. The exclusion of these random effects from the final estimations enables the model to provide outputs that can be consistently compared across teams, regardless of their baseline strength while accounting for the repeated structure of the dataset. Built using spatio-temporal data derived from major competitions at elite standards, the model highlights teams with strong performances during this period, identifying them as top performers across various aspects of the game. As such, xGV provides valuable information to coaches and analysts and can help refine tactical decision-making and arguably represents a significant advance in rugby analysis by quantifying the value of the game at any given moment, enabling objective comparisons across teams and strategies.

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Expected game value; general additive mixed model; performance analysis; rugby union; sports analytics

1. Introduction

A key aim of performance analysis is to provide objective information to coaches or players about specific actions or choices made, based upon the available options and the specific circumstances of the game (Nevill et al., 2008). Over recent years, the importance of data analysis in performance assessment and decision-making in sport generally (Baca et al., 2020; O'Donoghue & Holmes, 2014) is ever increasing. Traditionally, understanding how players and teams perform is often achieved through determining and analysing key performance indicators (KPI). In rugby, the

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identification of KPI has been the subject of numerous publications (Bremner et al., 2013; Hughes et al., 2017; Jones et al., 2008; Kvasnytsya et al., 2024; Scott et al., 2023; Sella et al., 2019; Watson et al., 2017). However, many limitations can be identified in these studies, which often lack context, and where certain choices of descriptive statistics might be questionable. These issues also illustrate the gap between performance-based research and the practical needs of coaching practitioners (Colomer et al., 2020; Watson et al., 2021).

Traditional methods using simple counts of game events are useful for breaking down performance, identification and counts of typical KPI, such as line breaks or turnovers. However, these might not provide actionable insights for coaches and players in developing their tactics. Moreover, not all events are equal, and context is crucial: an event occurring several metres from the try line will likely have a different impact compared to one in the middle of the field (McGarry, 2009). Thus, there is a need to weigh each type of action notably according to its pitch position. The use of spatio-temporal data obtained through player tracking systems has enabled the emergence of more complex analytical metrics to assist in the evaluation of match performance.

As early as 1970, Carter and Machol introduced the concept of expected goals to quantify the expected score of a possession (Carter & Machol, 1971). This notion was further explored by Carroll et al. (1988) and extended to other team sports including baseball (Albert & Bennett, 2001; Lindsey, 1963) and hockey (Macdonald, 2012), and football where it is defined based upon attempted shots at goal (Green, 2012). More recently, in rugby union, commercial match data providers like StatsPerform have also proposed expected points models built from the start of each possession, using several characteristics and primarily field position (StatsPerform, 2023). These models, similar to expected goals in football, aim to provide an expected score based upon created scoring opportunities and enable assessment of match performance beyond the actual score. However, rugby union involves possession sequences that can last several phases and contain numerous events. As such, these methods do not have sufficient granularity to allow assessment of the impact of the choice of play over the course of the attacking and/or defending phase. Additional advanced metrics, generally referred to as expected possession value (EPV) models, have therefore been developed across a range of sports. In this article, we propose a method called expected Game Value (xGV) for rugby union, which assigns a quantitative value to each game state for both teams. This type of methodology has yet to be applied in a sport like rugby, offering the potential for various analyses. While metrics with similar objectives are being developed in other sports, they do have limitations that we have considered in this new approach.

Existing models have their own features to quantify more continuously the impact of each game action for evaluating team or player performance (Baumer et al., 2023). Some of these models use stochastic methods using Markovian principles (Howard, 1960), as seen in basketball (Cervone et al., 2016), ice hockey (Liu & Schulte, 2018; Routley & Schulte, 2015), football (Fernández et al., 2021; Rudd et al., 2011; Singh, 2019; Yam, 2019), and rugby league (Kempton et al., 2016; Sawczuk et al., 2022). In rugby league, which shares similarities with rugby union, these methodologies discretise the field into zones to estimate the transition from each zone, which can be a limitation when trying to assess territory gained or lost. Being able to account for the continuous aspect of field location seems crucial in a territorial invasion sport like rugby union.

A second approach to expected possession value models is to use learning models through regression or machine learning methods to describe a possession state using various contextual elements to model a target score variable. For example, Decroos et al. developed the Valuing Actions by Estimating Probabilities (VAEP) method, which uses a CatBoost algorithm to estimate the probability of scoring and conceding points in the next N events, considering the previous three actions for each state (Decroos et al., 2019). XGBoost algorithms have also been employed in the Possession Value Framework proposed by StatsPerform, to estimate the probability of scoring within the next ten seconds (StatsPerform, 2020), or in the Goals Added method (Kullowatz, 2020), which models expected goals over two possessions, using characteristics such as field position, action type, and possession length up to the event. The commercial football data provider StatsBomb has also developed a method called On-Ball Value (OBV) (StatsBomb, 2021), which separately models expected points for the current possession and the opponent's next possession using a gradient-boosted decision tree model. It includes features such as field position, action type, and whether there is defensive pressure.

All these methods rely on machine learning algorithms and do not account for the dependency structure of the data, as they overlook the repeated measurements by the team. This limitation can lead to biased estimations, as the conclusions may be disproportionately influenced by the representation of certain teams within the dataset (Brill et al., 2024; Davis & Robberechts, 2024). These machine learning methods are also often subject to over-fitting problems (Brill et al., 2024). The use of statistical methods rather than machine learning may be preferable to resolve these limitations.

In rugby union, to our knowledge, no comparable expected possession value models have been proposed in the scientific literature to date. Developing such a method requires adapting the model to the unique characteristics of rugby, as the sport features multiple scoring methods with varying point values, unlike many other sports that model a single, often binary, outcome. Fitzpatrick and Nolan (2024) recently outlined several potential applications of an expected points method in rugby, such as comparing expected and actual values for goal kicks, in-play actions, or strategic decisions, but did not propose a method to compute these expected points. The inherent sequential nature of rugby, divided into phases of play by ruck events, makes it a sport that is fittingly structured and well suited to methodologies like expected Game Value.

The structure and aims of this article are as follows: the first section presents the methodology, starting with the time windows used and the scoring system for actions, followed by a detailed explanation of the modelling approach and the metrics derived from the estimations. The second section presents the results, showcasing the outcomes of these estimations and providing illustrative examples of how these metrics can be applied to evaluate team performance.

2. Materials and methods

2.1. Dataset description

The present dataset comprised elite competitions from the French Top 14, Gallagher Premiership, Super Rugby Pacific, United Rugby Championship, Investec Champions Cup, EPCR Challenge Cup, and international tournaments such as the Six Nations, The

Rugby Championship, Rugby Union World Cup, and summer and autumn test matches. Matches involving international teams outside the top 14 of the World Rugby rankings, as of 7 August 2022, were excluded. The study spans the 2022/2023 and 2023/2024 seasons, covering a total of 1416 matches, including 117 international matches, involving 72 teams (including 14 international teams) and 3065 professional players, of whom 717 are internationals. Data were sourced from the company StatsPerform and accessed through their API (certification ISO 9001:2015 n°FS 771,659 for data processing and analysing for sports). Spatio-temporal data of game events (ruck, lineout, scrum, tackle, receipts, etc.) were utilised and the coordinates of each event were encoded in a coordinate (X, Y) system, where X ranges from 0 to 1000 and Y from 0 to 700. The X-axis represents the field from try line to try line, while the Y-axis covers the width of the field from sideline to sideline.

3. Methodology

The objective is to estimate the expected points for and against at any given moment in the game, whether for the attacking or defending team. Game states are defined as kick-offs, lineouts, scrums, rucks, and turnovers. For each state, we estimate separately the chances of scoring and the risk of conceding points for the team in possession of the ball based on contextual factors of the action. We aim to calculate, at each game moment t, an expected game value for (xGVF) and an expected game value against (xGVA). The difference between these values gives the expected game value difference (xGVD), which we can more simply name *expected game value*.

3.1. Event time window

To account for both the scoring chances and the risk of conceding points, we consider as the time window, the sequence of possession following the modelled action and the opponent's next possession. A possession sequence is defined broadly as the total time until the opposing team regains the ball and not necessarily the end of a game phase. For example, if a team retains possession, and the opponent commits a defensive penalty, leading to a lineout, the subsequent play from the lineout is still deemed part of the same possession. Rucks within advantage plays following an infringement are excluded from the model.

3.2. Event time window scoring

Prior to modelling, a score for and against is developed for each event, equal to the number of potential points scored or conceded following the event within the current and next possession. The three ways to score points are through tries, drop goals, and penalties. The number of points from these actions depends on successfully kicking the ball through the posts, which is influenced by the skill of the kicker. However, this success probability is independent of the sequence of actions leading to the scoring opportunity. Therefore, we estimate the number of points without considering the observed success of the kicker. A mixed logistic model was used to determine the probability of converting a

penalty or a conversion based on the distance and angle from the ball to the posts, with random effects for individual players. The value of a try is thus:

$$Score_{try} = 5 + 2 \times P(conversion = 1|d_1, \alpha)$$

where d_1 and α represent the distance and angle between the conversion attempt and the posts.

A second mixed model was created to estimate the probability of a successful drop goal, incorporating variables such as the angle, the distance to the posts, and the distance between the pass of the player preceding the drop attempt and the reception of the ball by the player who kicks it (d_2), representing the estimated pressure on the player. In cases where there was no pass, the average value of the variable was imputed. The score for a drop goal is:

$$Score_{drop} = 3 \times P(conversion = 1|d_1, d_2, \alpha)$$

Another scoring scenario arises from a penalty, offering multiple options to the team. In this situation, a penalty kick can be chosen to try to obtain three points, the ball can be kicked directly into touch to obtain a lineout and retain possession, the scrum option can be selected, or a tap-and-go can be taken. For the second option, a linear mixed model was implemented to estimate the lineout position the team could achieve. Random effects for the penalty kicker were included, along with fixed effects for the field coordinate along the length, the distance from the penalty to the sideline, and whether it occurred on the left or right side of the field. To estimate the value of this lineout, we performed a first iteration of the models for lineouts (presented in the following section), considering only the probability of a successful penalty kick. From the estimated lineout position and its game value, the expected value of the scrum and the probability of converting a penalty kick, the value of a penalty can be written as:

$$\begin{aligned} Score_{penalty} = max(&3 \times P(conversion = 1|d, \alpha) \\ &- (P(conversion = 1|d, \alpha) \times \overline{Scorefor(s = start_{50})} \\ +(&1 - P(conversion = 1|d, \alpha)) \\ &\times \overline{Scorefor(s = start_{22})}, xGVD_{s=Lineout}(\hat{x}, y, \delta), xGVD_{s=Scrum}(x, y, \delta)) \end{aligned}$$

Where $xGVD_{s=Lineout}$ and $xGVD_{s=Scrum}$ are the difference between the expected game value for and against for lineout and scrum, respectively, based on the estimated lineout position $\hat{x}$ or the scrum position x and the player differential δ between the teams. Similarly, for penalty kicks, we subtract the risk of the next restart, represented by the average score for values, depending on whether it is a 22-metres or halfway restart, according to the penalty conversion probability.

3.3. Modelling of game states

The xGVF and xGVA values for each game state were determined using generalised additive mixed models (GAMM). The attacking team and defending team were included as random effects to account for repeated data structures but were not used for predictions since the aim is to model the value of a game state independently of team strength. Variable selection and model quality were evaluated through 5-fold cross-validation, and

metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Adjusted R^2 were calculated. All analyses were performed using R v4.3.1, with models implemented using the lme4 package v1.1–34 and the scam package v1.2–17.

For kick-off situations, where the model's performance was deemed insufficient and scenarios were largely similar, the average observed points for these start and restart sequences were used.

For all models of each type of action, penalised splines (P-splines) were used on the covariate representing the position of the action along the X-axis (length of the field). A positive monotonic constraint was added for the xGV-F and a negative monotonic constraint for the xGV-A to this same covariate. The other covariates used to model the values for lineouts include the side of the field where it takes place and the difference in the number of players on the field, notably if one or more players have been penalised (sinbin, sending off). For scrums, the distance of the scrum from the sideline was added in addition to the variables used for lineouts. Finally, the models for rucks and turnovers include all the previously mentioned variables as well as the gain or loss of pitch relative to the previous ruck along the X-axis. For the states S=[lineout, scrum, ruck, turnover] the equation for the models takes the form:

$$Score_{od}(S = s) = \beta_0 + f(X_1) + \sum_{j=2}^{k} \beta_j X_j + b_o + b_d$$

Where $Score_{od}$ represents the points for or against the offence team o against the team d in defence, $f(X_1)$ is the P-spline function applied to the X-coordinate of the field, with a monotonic constraint, X_j, 2<k, are the covariates used for the different types of game states and b_o and b_d are the random effects specific to the team in possession of the ball and the team in defence. Based on these estimations for each game state and the scores previously presented, the expected game values for and against are then defined as shown in Table 1.

The expected game value difference (xGVD) at time t is defined as:

$$xGVD_t = xGVF_t - xGVA_t$$

For the defending team, its expected game value is therefore the opposite of that for the attacking team:

$$xGVD_{od} = -xGVD_{do}$$

Various indicators can be derived from this method. To quantify over- or under-performance, we can compare the observed values to the predictions represented by the expected models. This gives:

$$efficency_t = Score_t - xGVD_t$$

To evaluate the short-term quality of an action, one option is to calculate the difference between the expected game value at time t and at time t + x:

$$\Delta xGVD_x = xGVD_{t+x} - xGVD_t$$

Where x is the minimum between the predefined game time to evaluate the outcome after the action, and the potential end of a play sequence.

Table 1. Definition of expected game value for (xGVF) and against (xGVA). The values assigned to actions where the notation $\overline{score}$ is used represent the average of the observations. For those represented by the notation score that are defined by the score estimation from generalised additive mixed models (GAMM). The probabilities are determined using generalised linear mixed models (GLMM). Random effects are not included in the predictions.

	xGVF	xGVA
Try	$5 + 2 \times P(conversion = 1)$	$\overline{Scorefor(s = start_{50})}$
Drop	$3 \times P(conversion = 1)$	$P(conversion = 1) \times \overline{Scorefor(s = start_{50})} + (1 - P(conversion = 1) \times \overline{Scorefor(s = start_{22})}$
Penalty	0	$max(3 \times P(conversion = 1) - \Big(P(conversion = 1) \times \overline{Scorefor(s = start_{50})} + (1 - P(conversion = 1) \times \overline{Scorefor(s = start_{22})}, xGVD(s = lineout), xGVD(s = scrum))$
$Start_{50,22,GL}$	$\overline{Scorefor\big(s = start_{50,22,GL}\big)}$	$\overline{Scoreagainst\big(s = start_{50,22,GL}\big)}$
Lineout	$\widehat{Scorefor(s = lineout)}$	$\widehat{Scoreagainst(s = lineout)}$
Scrum	$\widehat{Scorefor(s = scrum)}$	$\widehat{Scoreagainst(s = scrum)}$
Ruck	$\widehat{Scorefor(s = ruck)}$	$\widehat{Scoreagainst(s = ruck)}$
Turnover	$\widehat{Scoreagainst(s = turnover)}$	$\widehat{Scorefor(s = turnover)}$

4. Results

Figure 1 presents the xGVD values for the team in possession of the ball for distinct game states, with both teams having the same number of players on the field. Beyond the 40-metre line, a team in possession is in a positive situation, with a higher likelihood of scoring points than conceding them. Consequently, a team is also in a favourable position when defending within the opponent 40-metre area. In comparable field positions, set-piece phases such as lineouts and scrums have a higher expected game value than open play phases like rucks. The imbalance caused by a turnover also results in a higher game value than rucks in comparable positions.

In Figure 2, the analysis of teams showed that the Irish national team and Leinster Rugby have the best balance between opportunities and efficiency in both attack and defence. The English national team concedes few dangerous opportunities in defence, although their results on these occasions are rather average. Out of all the teams, New Zealand outperforms all other teams, both in attack and defence, relative to the chances they create. They are also the top-ranked team for their first-phase evaluation of lineouts and scrums (Table 2). The four best performing teams in open play attacks are Super Rugby clubs.

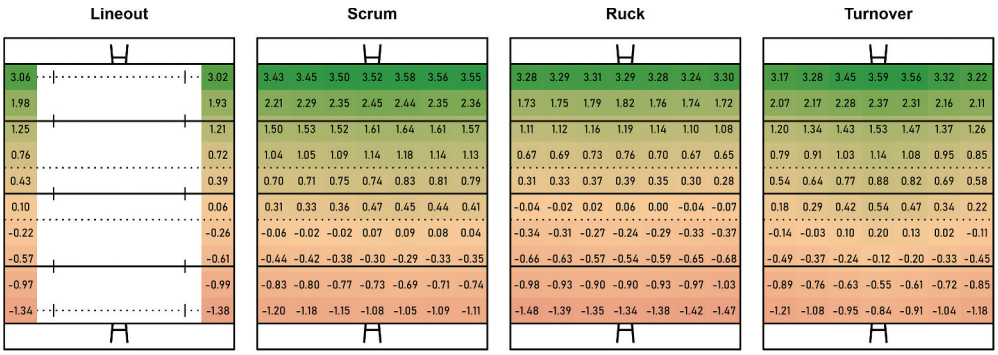

Figure 1. Expected game value difference (xGVD) of the team in possession for each action type depending on the pitch position.

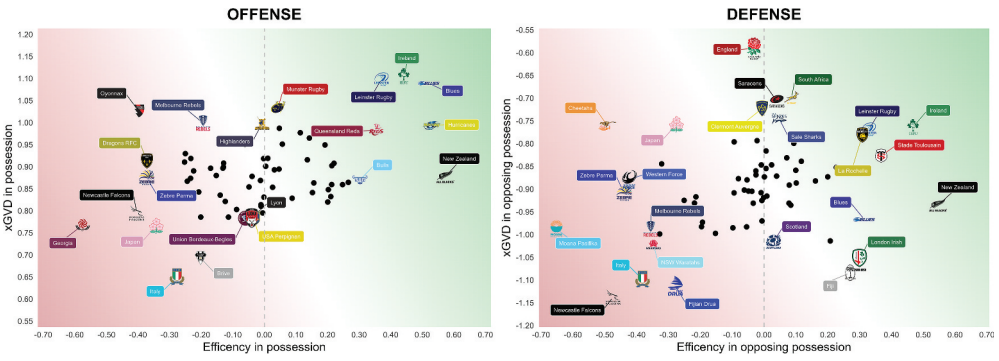

Figure 2. Scatter plot by team according to the average expected play value level (xGVD) and the average efficiency represented by the deviation from the conclusions obtained in attack and defence.

Table 2. Top 5 best teams in offence and defence on the average evolution of the expected game value on the following game phase ($\Delta xGVD_1$) according to the type of action.

	OFFENCE			DEFENCE		
Rank	Lineout	Scrum	Ruck	Lineout	Scrum	Ruck
1	New Zealand (+0.37)	New Zealand (+0.37)	Hurricanes (+0.19)	Glasgow Warriors (+0.10)	La Rochelle (+0.20)	Ireland (−0.01)
2	South Africa (+0.28)	Hurricanes (+0.31)	Chiefs (+0.18)	Crusaders (+0.08)	Hurricanes (+0.19)	Stade Toulousain (−0.02)
3	Queenslands Reds (+0.24)	Crusaders (+0.27)	Bulls (+0.17)	South Africa (+0.07)	Stormers (+0.17)	Chiefs (−0.04)
4	Bristol Bears (+0.23)	Chiefs (+0.26)	Blues (+0.16)	Leinster (+0.05)	New Zealand (+0.17)	Fiji (−0.04)
5	Bath Rugby (+0.23)	South Africa (+0.25)	France (+0.16)	La Rochelle (+0.04)	Stade Français (+0.17)	New Zealand (−0.05)

The expected Game Value for New Zealand increases on average by + 0.37 between their lineout and the subsequent phase of play.

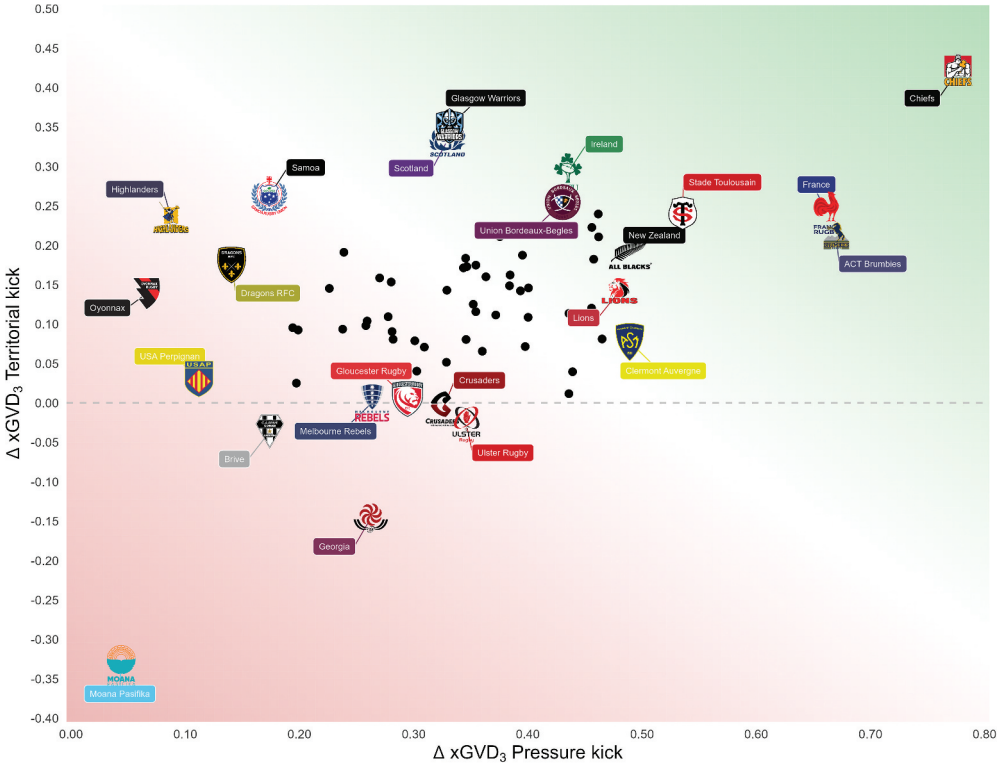

Figure 3. Scatter plot of the average difference between the expected game value before the pressure or territorial kick and the expected game value three phases after.

The Chiefs are by far the team with the best outcomes within the three phases following their kicks (Figure 3). Behind them, other teams that stand out for their pressure kicking include Brumbies and France, while territory kicking includes Glasgow Warriors and Scotland.

5. Discussion

The examples presented in this article illustrate the interest and possible uses of the expected game values model and its ability to highlight teams that have achieved significant results over a two-season period. For instance, the teams with the highest win percentages in our dataset are, in order, Ireland, Leinster, Blues, Chiefs, and France, all of which consistently and positively stand out across the different metrics analysed. Examining the relationship between our indicator and win percentage is one way to validate it, but assessing it more precisely remains challenging, as its purpose is to measure an aspect of performance that has not been quantified before (Davis et al., 2024). The expected game value method nevertheless offers numerous possibilities, particularly for performance evaluations and comparison of the choices made over the course of play.

One of the first applications here concerns the evaluation of over- or under-performance by comparing actual outcomes with expected ones (Fitzpatrick &

Nolan, 2024; Tippett, 2019). This type of analysis can be even more insightful when contextualised through different factors such as field zones, type of action, opponent, etc. Another application of the expected possession value method, as proposed in this article, is to evaluate game choices. For this purpose, the expected game value is compared before and after the actual choice is made. However, the question arises, over what period of time is it appropriate to evaluate the consequences of a game choice? Consideration of only the immediately following state, as classically done for some methods (Kullowatz, 2020; StatsBomb, 2021; StatsPerform, 2020), may be imperfect in rugby. In some situations, it can be judged that the consequences of a game choice, whether positive or negative, can occur in the following game phases. For example, a kick in play may not have a direct consequence when it is received, but it may disorganise the two teams and have an impact on the next phase of play. However, taking the value of an action that is too far away in time risks no longer evaluating the game choice but the events that will have taken place between the two. One possibility could be not to consider the value of a state at $t+x$, but instead to consider the average, maximum, or minimum, for example, of the xGV value from $t+1$ to $t+x$.

Set-piece phases such as lineouts and scrums are generally evaluated through a simple percentage of retained possession. To be more precise, some practitioners consider the next action or the outcome of the possession. However, it is difficult to compare actions with each other, especially based on their field position, and thus determine the level of success of these set-pieces. Using expected game value measures allows a more accurate quantification of a team's level of success during these phases of play. Moreover, there is simply no standard KPI to quantify the effectiveness of kicking plays or passing/movement combinations. xGV metrics now render possible the evaluation of all these tactical choices by quantifying their impact on the game.

A key advantage of this method, compared to those presented in other sports (Kullowatz, 2020; StatsBomb, 2021), is the implementation of a distinct model for each type of action. One advantage is that each model is customised with the parameters that best describe the context of the action. However, parameters that refer to the game choices we seek to evaluate should not be included in the models. For example, the number of players in the lineout should not be considered a contextual description but rather a decision to be evaluated based upon the lineout's location.

A key aspect of the methodology is the continuous consideration of field position, which, unlike other expected possession value models (Sawczuk et al., 2021), enables identification of the impact of each metre gained or lost, which is crucial in a territory-based sport like rugby union. Indeed, rugby is suited to this type of methodology owing to its sequential nature and the ease with which it can be broken down into distinct states. In general, open play is already often segmented at the ruck level into phases of play. Unlike other models, such as those in football where every action is accounted for (Decroos et al., 2019; Kullowatz, 2020; StatsBomb, 2021), in rugby, it seems more challenging to create models at a finer level than the ruck. One reason for this is that event-based data alone makes it difficult to sufficiently contextualise actions like a pass, a reception, or a kick, although this could become feasible with tracking data. Additionally, rugby is a sport of collective movements, where player positioning and passing are interdependent (Rodrigues & Passos, 2013). Thus, it seems more appropriate to consider

passing sequences as collective choices, which are then evaluated based on the preceding and following states.

A unique feature of the expected game value method proposed in this article is its ability to quantify both opportunity (via xGVF) and risk (via xGVA) for each action. Some expected possession value methods only measure offensive scoring chances (Rudd et al., 2011; Sawczuk et al., 2021; Singh, 2019). However, in rugby, given the consequences of turnovers or interceptions, it is also essential to consider risk. This is done by accounting for the next possession, as seen in OBV and G+ models (Kullowatz, 2020; StatsBomb, 2021). A particular aspect of rugby is that points can be conceded even during a team possession, such as from an offensive penalty which can lead to an opposing score. xGVF and xGVA thus account for points from both possessions combined.

Here, we chose to use statistical models for estimation rather than the machine learning methods commonly employed in similar settings (Baldwin, 2021; Decroos et al., 2019; Harish et al., 2023; Kullowatz, 2020). While machine learning methods like XGBoost generally have better predictive performance than the statistical models used here, they also demonstrate some limitations (Brill et al., 2024). These methods often overfit due to their flexibility, making them sensitive to certain artefacts. Another limitation of decision tree-based models is that they inherently split values at specific thresholds. For example, to model lineout situations we would have a slightly staircase relationship between the position on the field and the chances of scoring points. Generalised additive models provide a smoother progression across the field while maintaining significant flexibility with the use of splines. This method, which uses penalised splines, makes it possible to avoid the need to predefine the number of knots and their positions. Another limitation of machine learning methods is that they do not account for the data dependency structure (Brill et al., 2024; Davis & Robberechts, 2024), and particularly not the teams or opponents, thereby potentially biasing the estimates. The results of these models may thus represent a weighted average of over-represented team games or actions. Integrating these team dependencies as random effects in the generalised additive allows for a reliable estimation of the other parameters for an average team. Unlike expected points models, which may incorporate team level into their predictions (StatsPerform, 2023), expected possession value models have a different objective, where it is crucial to obtain estimates independent of the teams. Excluding teams' coefficients from the predictions enables expected game value comparisons between teams without considering their inherent level. Considering these teams, even if they are not included in the predictions, allows for more robust models than modelling with XGBoost (Table S.1 in the supplementary material section).

5.1. Limitations

Certain limitations must be considered when using a model like the one developed here. First, the models are built only from event data, so there is no information on the position of each player on the field at a given time, which can make the models less efficient. However, game states are relatively stable phases, and thus, the constructed variables, especially field position, play a major role in this type of model (supplementary figure). The use of variables such as the distance from the previous ruck allows consideration of potential defensive imbalances, aiming to approximate the opponent's positions. Video

tracking data, not currently available in rugby union, will present a great opportunity to obtain more precise estimates (Cervone et al., 2016; Fernández et al., 2021; Mallepalle et al., 2020).

A further limitation regarding the models developed to quantify the opportunity level of a penalty was that the possibility that the team might decide to play the ball by hand was not accounted for. The data label did not allow us to identify such choices, so we did not include this option in the models. Although this choice is less common compared to attempts at goal or lineout kicks, one might imagine that its value is not significantly higher than the other options.

Looking at the results and Table S.2 (in the supplementary material section), one can see that there may be specificities between different competitions although this was not the aim of the present study. Nevertheless, the dominance of Super Rugby teams regarding offensive metrics suggests that game values specific to this competition could differ from those observed in other tournaments. This is highlighted by the performance of Southern Hemisphere national teams, which exhibit similar offensive strengths. In contrast, competitions like the Top 14 demonstrate better defensive values, reflecting the unique characteristics and strategic priorities of each competition. These observations indicate that the distinctions are more related to the style of rugby played within each competition, rather than the level (club and national team). For the coaching staff, having expected game values from a model trained on less heterogeneous competitions could yield values closer to those of their team's environment.

Finally, rugby union is a sport that is constantly evolving, particularly with regularly changing rules. During the two seasons covered by this study, there were no rules that could have significantly changed the values of the game. However, the values from this type of methodology would need updating if new regulations were to affect attacking or defensive strategies.

6. Conclusion

This study presents a novel methodology for obtaining expected game values (xGV) in rugby union, providing a comprehensive framework for assessing team performance and decision-making. By quantifying the opportunity and risk associated with each game state, this approach goes beyond traditional metrics, offering a nuanced perspective on both offensive and defensive phases. The results demonstrate the potential of xGV as a valuable tool for analysing performance across various game situations, from set-pieces to open play. Using generalised additive mixed models rather than machine learning methods allows for flexible and understandable estimation of game value while considering the repeated structure of the data across teams. In conclusion, the expected game value statistic now makes it possible to assign values to every game situation in rugby union and could be a powerful tool for analysts and coaches by providing a foundation for the development of future research studies on rugby match performance.

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No potential conflict of interest was reported by the author(s).

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